



Seamless Knitting

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👤 Reviewers	Laurent Vincent Kishore Ganesan

1. Introduction

1.1 Purpose of the study

This document describes the life cycle inventory (LCI) model for seamless knitting as part of the Carbonfact Textile LCA Database. Seamless knitting represents a highly integrated production technology in which complete garments are produced on large-diameter circular knitting machines without subsequent cutting and sewing. The model is based on the Shima Seiki WHOLEGARMENT® and Santoni SM8 series machines, which represent the two principal industrial platforms for this technology.

The aim is to provide a transparent, traceable, and updatable foreground inventory for 1 kg of seamless knitted garment, consistent with the modelling conventions applied across all knitting documents in this database.

1.2 System boundary

The system boundary covers gate-to-gate seamless knitting: from yarn at the creel to finished garment at the mill gate. **Included:** direct electricity, indirect

energy, lubricant and waste mineral oil (inline auxiliary), capital goods and EoL (inline auxiliary), and building infrastructure. **Excluded:** fibre and yarn production, textile and material waste (handled externally via the IOR — no inline yarn-loss or textile-waste exchange in the dataset), finishing, packaging, and transport.

Functional unit: **1 kg of seamless knitted garment**. Background: ecoinvent 3.12, system model: cut-off by classification.

Note on sources: machine weight and energy range data are drawn from manufacturer datasheets (Shima Seiki, Santoni) and academic sources. Some supporting references (K&G Garment, Rakib Ahmed thesis) are commercial or grey literature and are used only for indicative benchmarking.

2. Process Description

Seamless knitting on large-diameter circular flat-bed machines (body-size machines such as the Santoni SM8-TR1S) produces complete garment tubes in a single operation, eliminating the cut-and-sew waste that typically accounts for 10–25% of fabric mass in conventional garment production (GIZ, 2023). Achieved material utilisation exceeds 95%, corresponding to less than 5% total material loss (K&G Garment, 2025).

However, seamless knitting is more energy-intensive per kilogram of output than conventional circular fabric knitting, because of complex knitting paths, longer cycle times, and higher frictional resistance — particularly with fine yarns. Installed power is 1.5–2.2 kW per machine, and typical productivity is 0.7–2.0 garments per hour. Based on EcoBalyse (ADEME, 2024), Maity et al. (2021), and Shima Seiki datasheets, the specific electricity consumption for seamless knitting is modelled at **1.5 kWh/kg** — the mid-range of the empirical 1.2–2.1 kWh/kg envelope.

The reference machine for capital goods allocation is the **Santoni SM8-TR1S**, a widely used large-diameter seamless machine with a published mass of 750 kg (Santoni, 2023) and a productivity of 26.2 kg/h (Textileworld.com, 2007; Maity et al., 2021).

3. Life Cycle Inventory

The inventory below represents the average scenario. Yarn losses and waste-yarn treatment are excluded and handled by the caller via the IOR (Section 5). Only auxiliary waste — lubricant and machine EoL — is modelled inline.

Flow	Amount	Unit	Dataset	
Technosphere inputs				
Lubricant, average	0.008	kg	`lubricating oil production {RoW} \	cut-off` — Sandin et al. (2019), p. 45: upper-bound 8 g/kg for high-load operation
Waste mineral oil (lubricant EoL)	0.008	kg	`market for waste mineral oil {RoW} \	cut-off`
Electricity				
Machine electricity (direct)	1.5	kWh	`market group for electricity, medium voltage {GLO} \	cut-off` — mid-range of 1.2–2.1 kWh/kg (EcoBalyse, ADEME 2024; Maity et al., 2021)
Indirect energy (HVAC, compressors, lighting, cleaning)	4.46	kWh	`market group for electricity, medium voltage {GLO} \	cut-off` — shared default: $2.4 \div 0.35 \times 0.65$
Capital goods				
Building machine (capital good)	5.33×10^{-7}	unit	`market for building machine {GLO} \	cut-off` — $750 \text{ kg} \div 1,886,400 \text{ kg} \div 2,000 \text{ kg/unit}$
Building infrastructure	1.61×10^{-5}	m ²	`market for building, hall {GLO} \	cut-off`

Flow	Amount	Unit	Dataset	
Waste flows (auxiliary only)				
EoL machine — steel (85%)	3.38×10^{-4}	kg	`market for waste steel {RoW} \	cut-off` — 637.5 kg ÷ 1,886,400 kg
EoL machine — other (15%)	5.96×10^{-5}	kg	`treatment of waste electric and electronic equipment, shredding {GLO} \	cut-off`

Note: the foreground dataset does not include yarn input or textile-waste flows. Yarn losses (2.5% default) are handled externally via the IOR by the caller.

4. Capital Goods and Building Infrastructure

Shared methodology: Capital Goods & End-of-Life (Machinery) in LCA — Full modelling framework and default values. This section states only seamless-knitting-specific inputs.

Shared methodology: Building Infrastructure in LCA — Full allocation framework. Section 4.2 states only the seamless-knitting-specific value.

4.1 Machine parameters

Parameter	Value	Source / note
Machine model	Santoni SM8-TR1S	Santoni (2023), SM8-TR1S datasheet
Machine mass (kg)	750	Santoni (2023)
Service life (years)	15	Average scenario
Operating hours per day	16	Average scenario
Operating days per year	300	Average scenario

Parameter	Value	Source / note
Productivity (kg/h)	26.2	Textileworld.com (2007); Maity et al. (2021)
Lifetime output Q (kg)	1,886,400	$15 \times 16 \times 300 \times 26.2$
Machine allocation (kg/kg garment)	3.97×10^{-4}	$750 \div 1,886,400$
ecoinvent units ($\div 2000$ kg/unit)	5.33×10^{-7}	$3.97 \times 10^{-4} \div 2,000 \times (750/750)$

Machine EoL: 85% steel (637.5 kg) → **3.38×10^{-4} kg/kg garment** → market for waste steel {RoW} | cut-off; 15% other (112.5 kg) → **5.96×10^{-5} kg/kg garment** → treatment of waste electric and electronic equipment, shredding {GL0} | cut-off.

Note on rounding: Summary values of 0.0003 kg/kg (steel) and 0.0001 kg/kg (other) are conservative rounded defaults. Use the precise figures above for recalculation.

4.2 Building infrastructure

Building infrastructure is allocated using the Santoni SM8-TR1S envelope (2,700 mm × 3,150 mm ≈ 8.5 m²) × cell factor 3–4×, yielding A_{alloc} in the range 25–34 m². The adopted default of **1.61×10^{-5} m²/kg** is cross-validated against the openLCA Nexus organic cotton sweater benchmark (GreenDelta, 1.61×10^{-5} m²/kg for seamless knitting). Seamless knitting has an intermediate building intensity between circular fabric knitting (lowest) and flat or 3D knitting (highest).

ecoinvent activity: market for building, hall {GL0} | cut-off (unit: m²).

4.3 Indirect energy

Shared methodology: Indirect Energy in LCA — Shared Methodology (Knitting)
— Full derivation, evidence base and subsystem distribution.

Indirect energy is modelled at **4.46 kWh/kg**, consistent with all knitting processes in this database (ADEME, 2024: $2.4 \div 0.35 \times 0.65 = 4.46$ kWh/kg). The subsystem distribution and evidence base are documented in the shared methodology page.

5. Input–Output Ratio (IOR)

Definition: $IOR = 1 / (1 - L)$, where L is the yarn loss rate (as a fraction). For every 1 kg of finished seamless knitted garment, the upstream yarn input must be scaled by IOR.

How the loss rate is derived

Seamless knitting eliminates most cutting operations, achieving >95% material utilisation. The **2.5% default loss rate** reflects yarn breaks, machine setup waste, and start-up trials. Sources: Maity et al. (2021); Rakib Ahmed (2024, NTNU thesis). The narrow range of material losses for this technology means a single default value is applied without BCS/WCS scenarios.

IOR value

Scenario	Loss (%)	IOR (kg/kg)	Input required (kg)	Loss per kg output (kg)
Default	2.5%	1.0256	1.0256	0.0256

How to apply: for 1 kg of seamless knitted garment, multiply upstream yarn input by **1.0256**. The 0.0256 kg of lost yarn is routed by the caller to `market for waste yarn and waste textile {RoW} | cut-off`. No inline yarn-loss or textile-waste exchange is included in the dataset itself.

6. Inferences and Limitations

The direct electricity value (1.5 kWh/kg) is a mid-range estimate anchored to published manufacturer data and academic benchmarks. The empirical envelope of 1.2–2.1 kWh/kg reflects variation in machine efficiency, gauge, and garment complexity; primary mill measurements would narrow this range.

The indirect energy estimate (4.46 kWh/kg) is the shared default across all knitting technologies and is the dominant contributor to process GWP. For seamless knitting, where machine cycles are longer and auxiliary system loads

may be proportionally higher than in conventional circular knitting, sub-metered data would be particularly valuable.

Capital goods contribute less than 1% of process GWP. The machine allocation and EoL values are derived from the Santoni SM8-TR1S datasheet (Santoni, 2023) and are considered representative of large-diameter seamless machines.

7. References

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